Problem 10.18

A fund earns interest at a force of interest of

$$\delta(t) = 0.01t.$$

Calculate the effective rate of interest in the 10th year.

Solution

The 10th year is the interval from time 9 to time 10. For a variable force of interest, the accumulation factor over this interval is

$$1 + i_{10} = \exp\left\{\int_9^{10} \delta(t)\,dt\right\}.$$

Substituting $\delta(t) = 0.01t$ gives

$$1 + i_{10} = \exp\left\{\int_9^{10} 0.01t\,dt\right\}.$$

Evaluating the integral,

$$\begin{aligned}
\int_9^{10} 0.01t\,dt &= \left.\frac{0.01t^2}{2}\right|_9^{10} \\
&= \frac{0.01}{2}\left(10^2 - 9^2\right) \\
&= 0.005(100 - 81) \\
&= 0.095.
\end{aligned}$$

Therefore,

$$\begin{aligned}
i_{10} &= e^{0.095} - 1 \\
&\approx 0.09965885513.
\end{aligned}$$

Hence the effective rate of interest in the 10th year is

$$\boxed{i_{10} \approx 9.9659\%}.$$